THE COLLISION SPECTRUM

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Abstract. For a prime base b and primitive odd Dirichlet character χ modulo b^2, the collision transform coefficient $\hat{S}^\circ(\chi)$ admits an exact factorization:

$$\hat{S}^\circ(\chi) = -\frac{B_{1,\overline{\chi}} \cdot \overline{S_G(\chi)}}{\phi(b^2)},$$

where $B_{1,\overline{\chi}}$ is the generalized first Bernoulli number and $S_G(\chi)$ is the diagonal character sum. By the standard Bernoulli–L-value formula, $|B_1| = (b/\pi)\,|L(1,\chi)|$, so the collision invariant's Fourier spectrum encodes L-function special values.

A Parseval identity gives an exact formula for the weighted second moment $\sum |L(1,\chi)|^2 \cdot |S_G(\chi)|^2$ in terms of the collision invariant's values on the finite group. The digit function computes this L-value moment exactly. Under a conditional zero-free hypothesis, the triangle inequality yields a separate bound connecting $L(1)$ to $L(s)$ for s in the critical strip.

At base 5, the factorization gives $|\hat{S}^\circ| \propto |L(1)|^2$ exactly. For quadratic characters in the family, the decomposition specializes to class-number data.

1. Introduction

The collision invariant S_ℓ, introduced in [4], is a function on the finite group $(\mathbb{Z}/m\mathbb{Z})^\times$ with $m = b^{\ell+1}$. The collision transform of [5] decomposes S_ℓ over Dirichlet characters. The antisymmetry theorem restricts the centered decomposition to odd characters, and the convergence theorem proves the resulting prime harmonic sum $F^\circ(1)$ is finite.

This paper identifies what the collision transform encodes. The collision spectrum is not merely correlated with L-function values. It is built from them. The main result is the decomposition theorem: each Fourier coefficient factors into a generalized Bernoulli number (encoding $|L(1)|$) and a diagonal character sum. The proof uses the slice formula from [4], the classical Bernoulli identity for character sums over fractional parts, and the vanishing of coset sums for primitive characters.

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2. The Decomposition Theorem

Theorem 1 (Decomposition)**.** *Let b be prime, $m = b^2$, and χ a primitive odd character modulo m. Then*

$$\hat{S}^\circ(\chi) = -\frac{B_{1,\overline{\chi}} \cdot \overline{S_G(\chi)}}{\phi(m)},$$

where $B_{1,\overline{\chi}} = (1/m)\sum_a a\,\overline{\chi}(a)$ and $S_G(\chi) = \sum_{n\in G}[\overline{\chi}(n{+}1) - \overline{\chi}(n)]$, and $G = \{n \in \{0,\ldots,m{-}1\} : \lfloor n/b \rfloor = n \bmod b\}$ is the diagonal set [4] *(elements whose base-b digits coincide), with $|G| = b$. Dirichlet characters are extended by $\chi(a) = 0$ for $\gcd(a,m) > 1$.*

Proof. Write $\phi = \phi(m)$. Expand $\phi\,\hat{S}^\circ = \sum_a S^\circ(a)\,\overline{\chi}(a)$ using the slice formula [4]: $S(a) = -1 - \lfloor a/b \rfloor + \sum_{n\in G} d_n(a)$ where $d_n(a) = \lfloor (n{+}1)a/m \rfloor - \lfloor na/m \rfloor$.

Step 1 (centering vanishes). The class mean $\overline{S}_R$ is constant on each spectral class $R = (a{-}1) \bmod b$, and the sum of $\overline{\chi}(a)$ over each class equals $\overline{\chi}(k)\sum_{u\in 1+b\mathbb{Z}/b^2\mathbb{Z}}\overline{\chi}(u) = 0$, because χ restricted to the subgroup $\{1{+}jb\}$ is non-trivial for primitive χ.

Step 2 (constant and floor terms). $\sum_a(-1)\,\overline{\chi}(a) = 0$. The fractional part $\{a/b\}$ depends only on $a \bmod b$, so

$$\sum_a \overline{\chi}(a)\,\{a/b\} = \sum_{k=1}^{b-1}\frac{k}{b}\sum_{a\equiv k \bmod b}\overline{\chi}(a) = 0,$$

since each coset sum vanishes (Step 1). Therefore $-\sum\lfloor a/b \rfloor\,\overline{\chi}(a) = -b\,B_{1,\overline{\chi}}$.

Step 3 (diagonal terms). The endpoints $n = 0$ and $n = m{-}1$ contribute nothing: $d_0(a) = 0$ (since $\lfloor a/m \rfloor = 0$), and $d_{m-1}(a) = 1$ for all units, giving $\sum_a \overline{\chi}(a) = 0$. For interior slices ($\gcd(n,m) = \gcd(n{+}1,m) = 1$), Lemma 2 gives

$$\sum_a d_n(a)\,\overline{\chi}(a) = [1 + \chi(n) - \chi(n{+}1)]\,B_{1,\overline{\chi}}.$$

Summing all $|G| = b$ slices: $\sum_{n\in G}\sum_a d_n(a)\,\overline{\chi}(a) = B_1[b - \overline{S_G(\chi)}]$.

Step 4 (combine). $\phi\,\hat{S}^\circ = 0 + (-bB_1) + B_1[b - \overline{S_G}] - 0 = -B_1 \cdot \overline{S_G}$. □

Lemma 2. *For primitive χ modulo m and $\gcd(n,m) = 1$:*

$$\sum_a \overline{\chi}(a)\,\{na/m\} = \chi(n)\,B_{1,\overline{\chi}}.$$

Proof. The substitution $a \mapsto n^{-1}a$ permutes the units and gives

$$\chi(n)\sum_a \overline{\chi}(a)\,\{a/m\} = \chi(n)\,B_{1,\overline{\chi}},$$

since $\{a/m\} = a/m$ for $1 \le a \le m-1$. $\square$

All identities involving F° and $P(s,\chi) = \sum_p \chi(p)/p^s$ are understood up to the finite contribution of primes $p \le m$, which is irrelevant for convergence.

3. The L-Encoding

The Bernoulli number is the L-function in disguise.

Corollary 3. $|\hat{S}^\circ(\chi)| = (b/\pi\phi)\,|L(1,\chi)| \cdot |S_G(\chi)|$.

Proof. By the generalized Bernoulli–L-value formula for odd primitive characters [1], $|B_{1,\overline{\chi}}| = (b/\pi)\,|L(1,\chi)|$. $\square$

Remark 4 (Connection to short partial sums). Computation verifies that for all primitive odd χ modulo b^2 with $b \le 13$,

$$|S_G(\chi)| = 2\Big|\sum_{k=1}^{b-1} \overline{\chi}(k)\Big|.$$

At base 5 (from the verified identity above), the partial sum further satisfies $|\sum_{k=1}^{4} \overline{\chi}(k)| = (\sqrt{5}/2)\,|B_{1,\overline{\chi}}|$ for all 8 primitive odd characters modulo 25, giving $|\hat{S}^\circ| \propto |L(1)|^2$. A proof of the general identity is open.

Remark 5. For quadratic odd primitive characters χ_D with fundamental discriminant $D < -4$, $|L(1,\chi_D)| = \pi h(D)/\sqrt{|D|}$. When such characters appear among the primitive odd characters modulo m, the decomposition theorem specializes to class-number data.

4. Moment Bounds

Theorem 6 (Moment identity)**.** *For prime b and $m = b^2$:*

$$\sum_{\substack{\chi \bmod m \\ \chi \text{ prim. odd}}} |L(1,\chi)|^2 \cdot |S_G(\chi)|^2 = \frac{\pi^2\phi(m)}{b^2} \sum_{\substack{a \bmod m \\ \gcd(a,m)=1}} |S^\circ(a)|^2.$$

Proof. Parseval's identity on $(\mathbb{Z}/m\mathbb{Z})^\times$ gives

$$\sum_\chi |\hat{S}^\circ(\chi)|^2 = \frac{1}{\phi}\sum_a |S^\circ(a)|^2.$$

By the antisymmetry theorem [5], $\hat{S}^\circ(\chi) = 0$ for even χ. For imprimitive odd χ induced from a character ψ modulo b, both $S_G(\chi) = 0$ and $\hat{S}^\circ(\chi) = 0$.

For S_G: since $G = \{r(b{+}1) : 0 \le r \le b{-}1\}$ and χ depends only on residues modulo b, we have $\overline{\chi}(r(b{+}1){+}1) = \overline{\psi}(r{+}1)$ and $\overline{\chi}(r(b{+}1)) = \overline{\psi}(r)$, so $S_G(\chi) = \sum_{r=0}^{b-1}[\overline{\psi}(r{+}1) - \overline{\psi}(r)] = 0$ (telescoping, using $\psi(0) = \psi(b) = 0$).

For $\hat{S}^\circ$: on each coset $\{a : a \equiv k \pmod b\}$, the imprimitive $\overline{\chi}$ is constant, while S° has mean zero on that coset by definition of centering. Each coset contributes zero, hence $\hat{S}^\circ(\chi) = 0$. Only primitive odd characters survive. Substituting the L-encoding (Corollary 3) and rearranging gives the identity. □

The right side is computable: S° is explicitly determined on the $\phi(m)$ units, and the weights $|S_G(\chi)|^2$ on the left are determined by the collision geometry.

Remark 7. At base 5, the identity $|S_G| = (5\sqrt{5}/\pi)\,|L(1)|$ (verified for all 8 primitive odd characters) converts the moment identity into $\sum |L(1,\chi)|^4 = c_5 \sum |S^\circ(a)|^2$: a fourth moment of L-function special values, computable from the collision invariant.

Corollary 8 (Conditional cross-moment)**.** *For prime b and $m = b^2$: if every primitive odd L-function modulo m satisfies $L(s,\chi) \neq 0$ for* $\mathrm{Re}(s) > \sigma_0$*, then for real $s > \sigma_0$ the weighted moment*

$$\frac{1}{\phi(m)} \sum_{\chi \text{ prim. odd}} |B_{1,\overline{\chi}}| \cdot |S_G(\chi)| \cdot |P_{>m}(s,\chi)|$$

is bounded below by $|F^\circ_{>m}(s)|$, where $P_{>m}(s,\chi) = \sum_{p>m} \chi(p)/p^s$ and $F^\circ_{>m}(s) = \sum_{p>m} S^\circ(p)/p^s$.

Proof. The zero-free hypothesis ensures that $P_{>m}(s,\chi)$ is well-defined for each primitive odd χ. Using the transform expansion from [5] and the vanishing of even and imprimitive odd coefficients established above, $F^\circ_{>m}(s) = \sum_\chi \hat{S}^\circ(\chi)\, P_{>m}(s,\chi)$ with only primitive odd characters contributing. The triangle inequality gives the bound. □

Remark 9. The bound relates $|L(1,\chi)|$ (from B_1, at the edge of the critical strip) to $|P(s,\chi)|$ (in the strip). Since $P = \log L - H$ and H is small for large b, this is approximately a bound involving $|\log L(s,\chi)|$.

5. The Correlation Decay

The decomposition theorem shows what the collision spectrum encodes. What remains is how faithfully. The partial sum $P = \sum_{k=1}^{b-1} \overline{\chi}(k)$ has an exact decomposition via the classical identity for short character

sums [2, 3]. After normalizing by the Gauss sum:

$$\widetilde{P} = L(1, \overline{\chi}) + \Delta(\chi),$$

where the packet

$$\Delta(\chi) = \frac{i}{\varphi(b)} \sum_{\substack{\xi \bmod b \\ \xi(-1)=1,\, \xi \neq 1}} \tau(\overline{\xi})\, L(1, \xi\overline{\chi})$$

is a sum of twisted L-values.

b	mean $\lvert\Delta\rvert/\lvert L\rvert$	std	std $\cdot \log b$
5	0.80	0.65	1.05
7	1.03	0.65	1.26
13	1.11	0.50	1.28
19	1.10	0.42	1.23
31	1.07	0.33	1.12
43	1.06	0.29	1.09

TABLE 1. The packet Δ appears to have constant magnitude relative to $L(1)$ (mean ≈ 1.08) and uniform phase ($\langle \cos\theta \rangle = 0.000$). The observed standard deviation of $|\Delta|/|L|$ is consistent with decay as $c/\log b$.

Computation using [6] suggests three properties of the packet. Its magnitude relative to $L(1)$ appears constant (mean $|\Delta|/|L| \approx 1.08$). Its phase relative to $L(1)$ appears uniformly distributed. And the standard deviation of the ratio $|\Delta|/|L|$ is consistent with decay as $c/\log b$ (Table 1).

This observed variance decay is consistent with the measured correlation between $|\widetilde{P}|$ and $|L(1)|$: at small b, the high variance of $|\Delta|/|L|$ allows $|\widetilde{P}| = |L + \Delta|$ to track $|L|$; at large b, the low variance makes $|\Delta|/|L|$ nearly constant, and the uniform phase washes out the L-specific information.

6. Remarks

The decomposition theorem explains the observations of the companion paper [5]: the anti-correlation between collision coefficients and prime character sum magnitudes, the absence of the principal-character term, and the mod-3 structure are all consequences of the factor B_1 in the identity $\hat{S}^\circ = -B_1 \cdot \overline{S_G}/\phi$.

The proof uses three classical ingredients (the slice formula, the Bernoulli identity, and the vanishing of primitive character sums over

cosets) applied to a new object (the collision invariant). The identity is exact for every primitive odd character at every prime base.

The analytic continuation from [5] combined with the decomposition gives $\mathcal{F}^\circ(s) = -(1/\phi)\sum B_1 \cdot \overline{S_G} \cdot [\log L(s,\chi) - H(s,\chi)]$. Near a zero ρ of $L(s,\chi_0)$, the coefficient of the divergent term involves $L(1,\chi_0)$ through B_1: the singularity at depth s is weighted by the health at $s = 1$.

What remains open is the proof of the variance decay $\mathrm{std}(|\Delta|/|L|) \sim c/\log b$ from the Apostol formula: this would establish the correlation decay between partial character sums and L-function special values as a theorem.

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